INCH-POUND

MIL-C-24643/23E
22 November 1994
SUPERSEDING
MIL-C-24643/23D
14 March 1994

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 300 VOLTS, TYPE LSTTSU
(INCLUDING VARIATION LSTTSA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24643.

REQUIREMENTS:

Qualification required.

Construction (watertight with circuit integrity)

First - Copper conductor, uncoated, AWG size 22, class K stranding.
Second - Extruded Silicone rubber insulation (see fourth). Minimum thickness 0.017 inch.
* Third - Insulation jacket of polyamide, per ASTM D4066 colored (see fourth) or transparent at manufacturer's option. Minimum average wall thickness 0.003 inch. Diameter over insulated and jacketed conductor shall be 0.076 inch minimum. Twisted pair identification code applied by method 6.
Fourth - Two conductors, one white, one black, twisted together to form a pair having a lay length not greater than
* 2-1/4 inches. For LSTTSU 1-1/2 only, three conductors, one white, one black, one red, twisted together to form a triad having a lay length not greater than 2-1/4 inches.
Fifth - The specified number of pairs (see table I) cabled together with a lay no greater than 10 inches. Cabling sequence shall be consecutive, starting with pair No. 1, from the center outward. Fillers shall be used as necessary to form a firm, well rounded assembly.
Sixth - Binder or combination binder and barrier.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Construction (watertight with circuit integrity) (continued)

Construction variant type LSTTSU

Seventh - Cross-linked polyolefin jacket. (see table I for thickness).

Construction variant type LSTTSA

Seventh - Cross-linked polyolefin jacket. (see table I for thickness).
Eighth - Braided metal armor

TABLE I. Details.

1/ Military part no. M24643/18	Size 2/	Types overall Diameter (maximum inches)		Number of pairs	Jacket thickness min avg, inch	Cold Bending mandrel diameter (maximum) (inches)
		LSTTSU	LSTTSA			
-01UN	1-1/2	3/.330	0.380	1-1/2	0.050	5
-02UN	3	.450	.500	3	.050	7
-03UN	5	.540	.590	5	.050	8
-04UN	10	.675	.725	10	.062	9
-05UN	15	.800	.850	15	.062	10
-06UN	20	.870	.920	20	.062	11
-07UN	30	1.080	1.130	30	.075	13
-08UN	40	1.200	1.250	40	.075	15
-09UN	50	1.400	1.450	50	.075	16
-10UN	60	1.450	1.500	60	.075	17

1/ Unarmored. When armor option or double shield is required, see MIL-C-24643 for configuration of part number
2/ Type LSTTSU and LSTTSA.
3/ LSTTSU 1-1/2 comprised of three conductors cabled together to form a triad.

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EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet at 25°C, maximum	18.60
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	2000
Insulation resistance - Megohms/1000 feet minimum	
Conductor to conductor	100
Conductor continuity	No failure
Group A:	
Visual and dimensional	No failure
Watertightness - see MIL-C-24643 for limits of water leakage	No failure
Crack resistance	No damage
Group B:	
Cross-linked proof test (percent, maximum)	
Insulation	50
Jacket (When tested at 200°C)	50
Gas flame 1/ - 1 hour (see note 1)	No failure
Drip - 95 ± 1°C	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Cold bending cable - see table I for mandrel diameter	No damage
Physicals (unaged)	
Insulation	
Tensile strength - lb/in^2, minimum	750
Elongation - percent, minimum	125
Jacket (cable)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Group C:	
Physicals (aged)	
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged minimum	60
Hot oil immersion	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Shrinkage	No failure
Heat distortion - percent of unaged, maximum	30
Permanence of printing (conductor) - cycles, minimum	25
* Permanence of printing (jacket) - cycles minimum	125
Cable sealant removability	No failure
Armor (type LSTTSA only) - conformance to material, construction and coverage	No failure

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EXAMINATION AND TESTS (contineud):	Requirements
Group D:	
Flame propagation (cable)	No failure

NOTE: 1. Braided armor, when present, shall be connected to neutral terminal of power supply. For gas flame test of LSTTSU 1-1/2, the three connectors shall be connected to phases A, B and C of the power supply. For test of all other sizes, connection of conductor pairs shall be in accordance with the following pattern:

(a) Pair #1: One conductor to phase A, one conductor to phase B
(b) Pair #2: One conductor to phase C, one conductor to phase A
(c) Pair #3: One conductor to phase B, one conductor to phase C
(d) Pair #4: One conductor to phase A, one conductor to phase B
(e) Pair #5: One conductor to phase C, one conductor to phase A
(f) Pair #6 through Pair #60: In similar sequence.

QUALIFICATION INSPECTION: Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:	Requirements
Cold working (minus 20 ± 2°)	No damage
Gas flame (3 hours) (see note 1)	No failure
Specific gravity of extruded silicone insulation, maximum	1.55
Aging and compatibility (cable) (125± 5°C)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content -percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	45
Insulation	35
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

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UNIT ORDERING LENGTHS:

	Type and Size	Feet (nominal)
*	LSTTSU/A-1-1/2 through -10	2500
*	LSTTSU/A-15 and -20	1500
*	LSTTSU/A-30 through -60	1000

NOTE: The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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